

GATI VARSHNEY

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Education

JSS Academy of Technical Education, Noida, India
Bachelor of Technology in Computer Science and Engineering (Data Science)

2023 – 2027
GPA: 8.7/10

Experience

Google Summer of Code 2026 Contributor, The Linux Foundation May 2026 – Aug 2026
AI-Driven Printer Compatibility & Recommendation Portal [\[Final Report\]](#)

- Designed an offline recommendation pipeline over OpenPrinting's Foomatic database of 6,657 printers, encoding each as a 463-dimension feature vector and ranking explainable matches by IDF-weighted cosine similarity, emitted as per-printer JSON shards so a page loads 6 KB instead of the 24 MB aggregate.
- Diagnosed that 86.8% of similarity scores saturated at a perfect 1.0, making ranking meaningless; added evidence damping and capability-conflict penalties that cut saturation to 0%, held in place by a reproducible evaluation harness that fails if any documented metric drifts.
- Engineered a deterministic client-side natural-language assistant over the same artifacts (printer search, driver lookup, comparison, contextual queries) that returns explicit data-gap answers rather than inventing capabilities; 384 tests over a 187-utterance corpus, no change to First Load JS.

Software Engineer Intern, Quin Que Inc. (Japan, Remote) Sep 2026

- Developed the operations console for Rapilogi, a multi-tenant warehouse reception platform: 6 screens covering tenant management, configuration, devices, users and audit logs, delivered across the OpenAPI contract, PostgreSQL migrations, Go handlers and the React frontend, with an MFA operator login.
- Uncovered a provisioning defect while reading design documents before implementing, generalizing the fix into a tenant-preset field rather than special-casing the tenant; implemented provisioning validation returning structured errors for missing configuration. 4 pull requests merged under project-leader review.

Open Source Contributor, OpenPrinting [\[Merged PRs\]](#) Nov 2025 – Aug 2026

- Architected a two-layer client-side search, live in production on the OpenPrinting site: a build-time AST pipeline (unified, remark-parse) compiles 200+ posts into a static JSON index, queried in-browser by MiniSearch with weighted relevance ranking and fuzzy matching, with no search server.
- Migrated 200+ Jekyll posts into Next.js with a Node.js script, handling frontmatter, media assets and redirects that preserve legacy URLs.
- Shipped the site's author system and homepage; 14 pull requests merged in total through maintainer review.
- Traced production 404s on trailing-slash URLs, including RSS links consumed by LWN, to a build config generated at deploy time that overrode the repository's; restored both URL forms across 7.5K+ routes and introduced a CI check that blocks any deploy with broken internal URLs.

Projects

cidx | Code Index for AI Coding Agents | Python, tree-sitter, SQLite, MCP [\[GitHub\]](#) 2026

- Built a zero-config local code index for AI coding agents: tree-sitter parses Python and TypeScript/JavaScript into a SQLite index of symbols and references, separating per-file extraction from whole-repository reference resolution; answers carry locations and signatures rather than file contents, each reference tagged by how it resolved.
- Defined and enforced a convergence invariant, incremental index identical to a cold rebuild, using content-hash invalidation, one SQLite transaction per file, and a debounced watcher with a reconciliation sweep for dropped OS events; verified by a Hypothesis property suite over random edit/delete/rename sequences plus a crash-injection test demonstrating no half-written index.
- Delivered five read-only MCP tools over stdio with token-budgeted responses, freshness stamps, and a grep fallback on every miss; validated on Django (3,000+ files, 75k symbols, 205k references, identical output on Linux and Windows) across a 3-OS × 3-Python CI matrix, published to PyPI.

Achievements

- Winter of Code 5.0:** Ranked Top 20 of 2800+ participants
- OpenPrinting Hall of Fame:** Credited for the author system, migration, site search, and homepage [\[Listing\]](#)
- LeetCode Knight:** Contest rating 1925, top 3.97% of contest participants globally [\[Profile\]](#)

Technical Skills

Languages: C++, Python, TypeScript, JavaScript, SQL

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs, OpenAPI, PostgreSQL, MongoDB, Mongoose, SQLite, JWT & NextAuth

AI Integration: Google Gemini API, Model Context Protocol (MCP), schema-constrained structured outputs (Zod)

Developer Tools: Git, GitHub Actions (CI/CD), Linux, tree-sitter, pytest, Hypothesis, Puppeteer, Postman

Core Subjects: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks